Predicting Patient Readmission within 30 Days

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Abstract

Unplanned hospital readmissions within 30 days of discharge represent a significant challenge in healthcare, affecting patient outcomes and increasing system costs. This study develops machine learning models to predict which patients are at elevated risk for readmission before discharge, enabling targeted interventions. Using data from both MIMIC-III and the UCI Diabetes datasets, we engineer features capturing temporal patterns, clinical activity, demographics, and medication information. We implement and compare multiple models including Logistic Regression, Random Forest, and XGBoost. Exploratory data analysis reveals that length of stay, previous hospitalization history, clinical complexity, and medication changes are associated with readmission risk. Despite computational constraints necessitating the use of data subsets, our findings demonstrate potential for clinical implementation through risk stratification, intervention targeting, and resource optimization. This work addresses an important healthcare challenge with direct applications to improving care transitions and reducing adverse outcomes.

1 Introduction

Unplanned hospital readmissions occurring within 30 days of discharge represent a critical issue in modern healthcare. They are not only detrimental to patient health and recovery trajectories but also place a considerable financial strain on healthcare systems globally. As noted in the project proposal, estimates suggest that nearly one-fifth of Medicare patients in the US experience readmission within this timeframe, translating into billions of dollars in potentially avoidable costs annually. These readmissions often signal gaps in care coordination, potential deficiencies in the initial treatment plan, inadequate patient preparation for self-care post-discharge, or unresolved underlying health issues.

The ability to accurately predict which patients are at an elevated risk for readmission *before* they leave the hospital is therefore of paramount importance. Such predictive capability empowers healthcare providers to proactively deploy targeted, patient-specific interventions. These interventions can range from intensified patient education and medication reconciliation to scheduling early follow-up appointments and arranging specialized home care services. By identifying these high-risk individuals early, healthcare institutions can optimize resource allocation, focusing intensive support where it's most needed. This proactive approach holds the potential to significantly reduce readmission rates, thereby improving patient safety and long-term health outcomes while simultaneously mitigating the substantial economic burden associated with unplanned returns to the hospital. This project was undertaken to develop and evaluate a machine learning model capable of providing such predictive insights using readily available patient clinical and demographic data.

2 Data Collection and Preprocessing Methods

2.1 Data Sources

The foundation of this project rests on two distinct, publicly accessible datasets, chosen to provide a breadth of clinical scenarios and patient populations:

1. **MIMIC-III (Medical Information Mart for Intensive Care III):** This comprehensive database offers a rich, longitudinal view of critical care patients. The specific tables utilized provided granular details:

    - DIAGNOSES_ICD: Contains patient diagnoses coded using the International Classification of Diseases (ICD) system, crucial for understanding patient comorbidities.
    - PROCEDURES_ICD: Lists procedures performed during the hospital stay, also coded using ICD, indicating interventions received.
    - ICUSTAYS: Details the timing and duration of Intensive Care Unit stays, a key indicator of illness severity.
    - LABEVENTS & D_LABITEMS: Provide results from various laboratory tests, offering insights into physiological status.
    - ADMISSIONS: Contains core information about each hospital admission, including admission and discharge times and types.
    - PATIENTS: Includes basic demographic data like gender and date of birth.

    Note: The analysis code utilized randomized subsets of these files (e.g., DIAGNOSES_ICD_random.csv), likely as a strategy to manage computational load during development and exploration. This implies that findings might need validation on the full dataset.

2. **UCI Machine Learning Repository: Diabetes 130-US Hospitals Dataset:** This dataset provides a focused view on the experiences of diabetic patients, a population often at higher risk for complications and readmissions. It includes demographic data, admission details (source, type), diagnostic codes (ICD-9), and specific information on diabetes medications and management.

2.2 Data Preprocessing and Feature Engineering

Transforming the raw data into a format suitable for machine learning required extensive preprocessing and the creation of informative features, primarily using the pandas and numpy libraries:

1. **Loading and Initial Cleaning:** Data was loaded from CSV files. Basic cleaning likely involved handling inconsistent data types or formatting, although specifics beyond date conversions are not detailed in the snippet.

2. **Target Variable Definition:** A consistent binary target variable, readmitted_30d, was crucial:

    - *MIMIC-III:* This required careful temporal calculation. For each admission, the time to the *next* admission for the same patient was computed. If this interval was $\leq$ 30 days, the flag was set to 1, otherwise 0. This captures the core definition of 30-day readmission.
    - *Diabetes Dataset:* The existing readmitted column (with categories '<30', '>30', 'NO') was directly mapped to the binary readmitted_30d flag (1 for '<30', 0 otherwise), simplifying the process for this dataset.

3. **Temporal Feature Engineering (MIMIC-III):** Extracting time-based insights was critical:

    - *Date Conversion:* Essential columns (ADMITTIME, DISCHTIME, DOB) were converted to datetime objects to enable calculations.
    - length_of_stay: Calculated as DISCHTIME - ADMITTIME, a fundamental indicator of resource utilization and potentially illness severity.
    - prev_admissions: Counting prior admissions provides a measure of a patient's history and potential chronic illness burden. Calculated using groupby('SUBJECT_ID') and cumcount().
    - days_since_last_discharge: The time gap between the current admission and the previous discharge for a patient. Shorter gaps often indicate unresolved issues or rapid deterioration.
    - age_at_admit: Calculated dynamically using DOB and ADMITTIME, providing a precise age relevant to the specific admission event, which is often more informative than a static age.

- admit_month: Extracted to explore potential seasonal patterns in admissions or readmissions.

4. **Clinical Activity Aggregation (MIMIC-III):** Summarizing clinical events per admission:
   - Counts of diagnoses (num_diagnoses), procedures (num_procedures), and distinct lab events (num_lab_events) were generated by grouping the respective tables by SUBJECT_ID and HADM_ID and counting entries. These serve as proxies for clinical complexity and intensity of care.
   - Total ICU stay duration (icu_los_days) was summed per admission from ICUSTAYS, indicating critical care needs.
   - These aggregated features were then merged back to the primary admissions DataFrame using SUBJECT_ID and HADM_ID.
5. **Demographic Feature Processing:** Patient gender and age were included as standard demographic predictors. For the Diabetes dataset, age ranges were mapped to numerical values (age_num).
6. **Medication Features (Diabetes Dataset):** Specific features were engineered to capture medication complexity and changes:
   - med_count: Summed the number of specific diabetes-related medications a patient was taking, reflecting the intensity of diabetes management.
   - med_change: A binary flag indicating if any change ('Ch') occurred in the diabetes medication regimen during the stay.
7. **Handling Missing Values:** The code shows fillna(0) applied after merging clinical activity counts/durations. This assumes that missing values in these contexts represent zero occurrences (e.g., no procedures performed). While practical, this might not be universally optimal. The proposal acknowledged missing data handling as a key step, suggesting more sophisticated techniques like mean/median/mode imputation or predictive imputation might have been considered or planned for other features.
8. **Data Integration:** Merging patient demographics (PATIENTS) with admission details (ADMISSIONS) in MIMIC-III using SUBJECT_ID created a unified record per admission.
9. **Normalization/Scaling:** Although not explicitly shown in the code snippet, the proposal mentioned normalization as a standard preprocessing step, which would typically involve scaling numerical features (e.g., using StandardScaler or MinMaxScaler) to prevent features with larger ranges from unduly influencing certain algorithms (like Logistic Regression or Neural Networks).

3 Insights from Exploratory Data Analysis

Exploratory Data Analysis (EDA) provided valuable insights into the relationships between various patient features and the likelihood of hospital readmission within 30 days.

3.1 Feature Correlation

Correlation heatmaps for both general (MIMIC) and diabetic patient subsets reveal several notable relationships (Figure 1). For the general MIMIC dataset, length of stay (LOS) and ICU days demonstrate a moderate positive correlation (0.37), indicating longer ICU stays typically result in extended hospitalizations. Additionally, number of procedures positively correlates with LOS (0.22) and lab events (0.32), suggesting higher complexity or severity cases. Conversely, previous admissions have a weak negative correlation (-0.14) with days since the last discharge, hinting that frequent readmissions tend to occur shortly after the previous discharge. For diabetic patients, time in hospital strongly correlates with the number of medications (0.47) and procedures (0.39), highlighting complexity and medication intensity in prolonged stays.

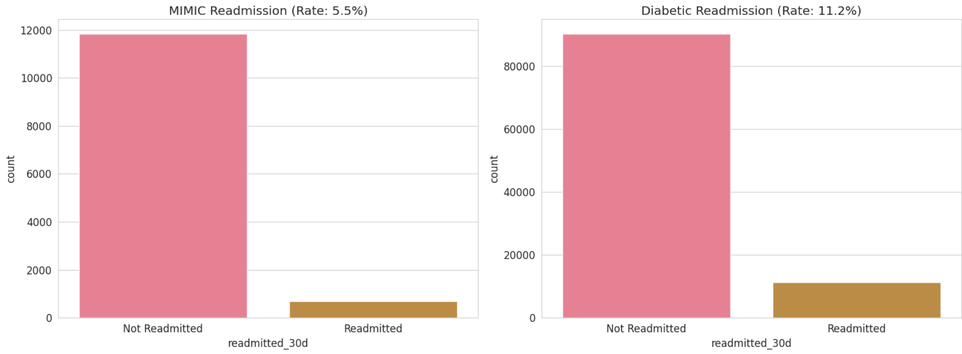

Figure 1: Feature Correlation Heatmaps for MIMIC and Diabetic datasets.

3.2 Medication and Insulin Impact

Medication analyses indicated that changes in medications are associated with higher readmission rates (Figure 2). Patients experiencing medication changes showed slightly higher readmission rates compared to those with no changes. Additionally, insulin status impacted readmission, with patients experiencing either significant increases or decreases in insulin having higher readmission rates compared to those with steady insulin levels or no insulin use.

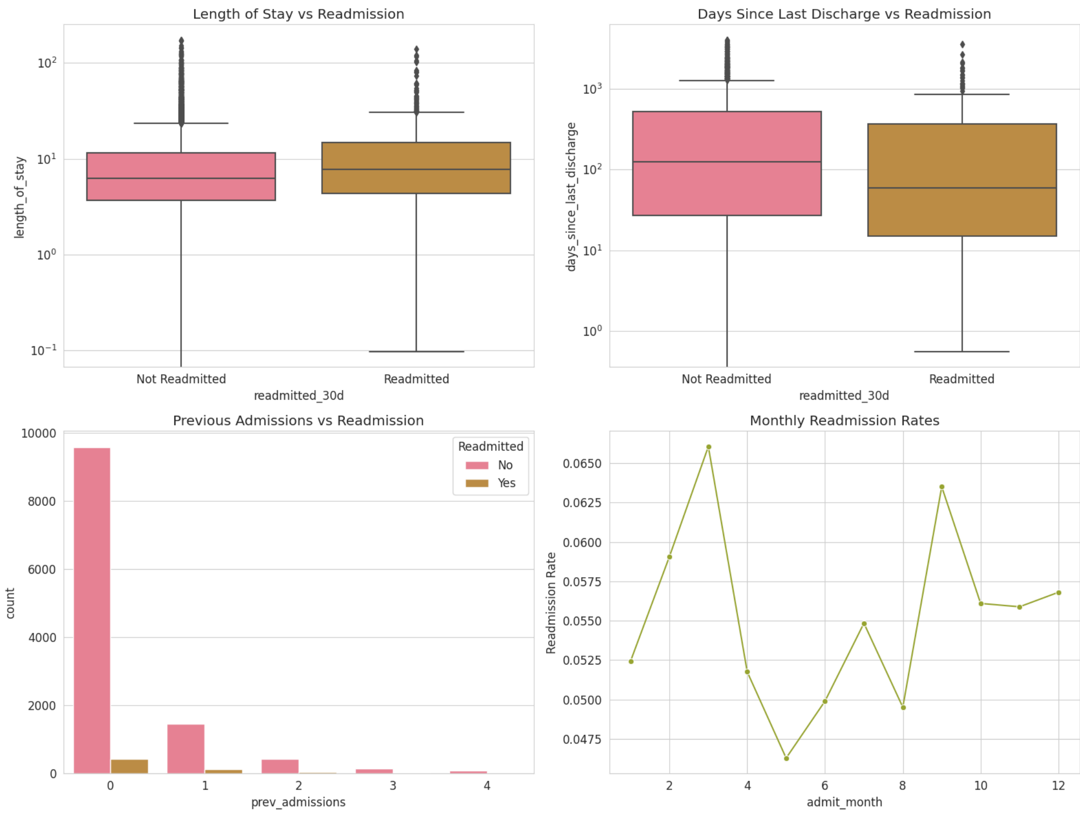

Figure 2: Medication and Insulin Status vs. Readmission Rate.

3.3 Demographic Factors

Demographic analyses revealed gender-specific insights and age influences (Figure 3). Female patients exhibited slightly higher readmission rates than males in both general and diabetic populations. The age distributions for readmitted and non-readmitted patients showed negligible differences, although slightly older median ages were observed among readmitted diabetic patients, indicating age as a subtle yet relevant factor.

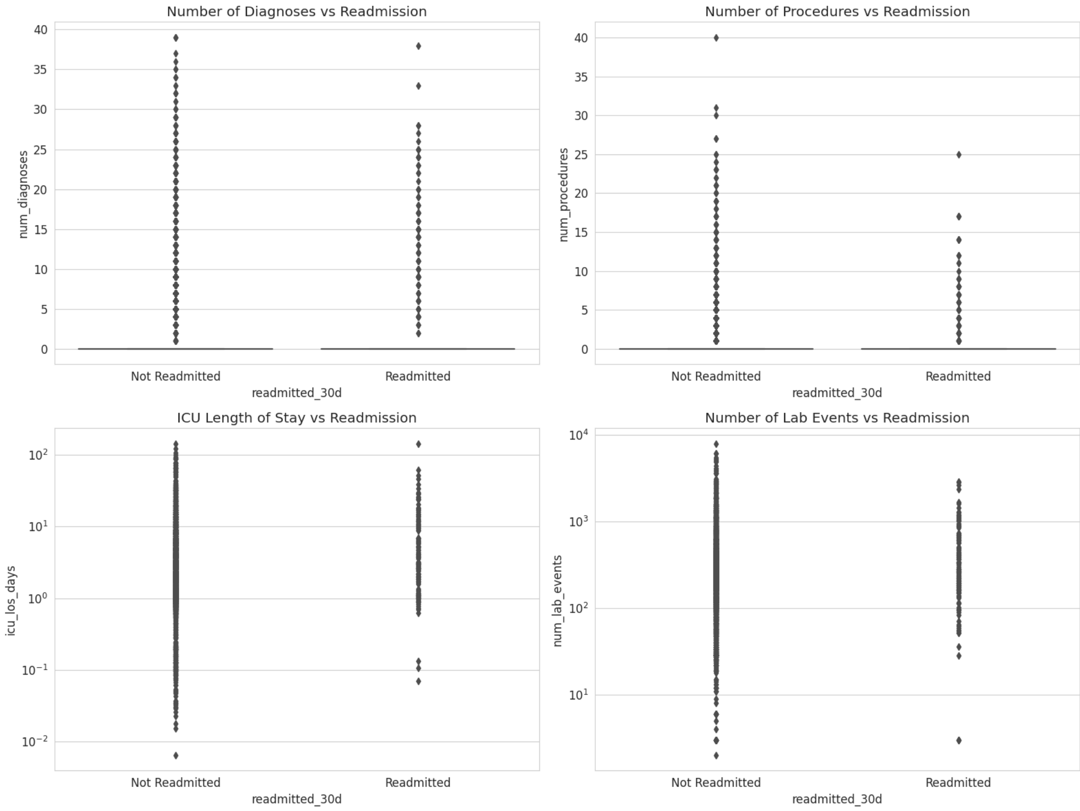

Figure 3: Age and Gender Influence on Readmission.

3.4 Clinical Complexity Indicators

Clinical complexity indicators such as the number of diagnoses, procedures, ICU length of stay, and laboratory events illustrated nuanced relationships with readmission (Figure 4). Higher numbers of procedures and diagnoses were commonly observed among readmitted patients, reflecting clinical complexity and increased healthcare utilization. Notably, readmitted patients frequently had longer ICU stays, reinforcing ICU duration as a significant readmission predictor.

3.5 Hospitalization Duration and Discharge Timing

Duration-related factors, specifically the hospital length of stay and days since last discharge, showcased intriguing patterns (Figure 5). Readmitted patients generally exhibited slightly longer median hospital stays and shorter periods since their last discharge. This observation underscores shorter recovery periods between admissions as critical risk indicators. Additionally, the monthly readmission rates indicated slight seasonal fluctuations, with peaks around early spring and autumn.

3.6 Overall Readmission Rates

Overall, the readmission rate for the general MIMIC population was approximately 5.5%, whereas diabetic patients had a notably higher rate of 11.2% (Figure 6). This significant disparity emphasizes diabetes as a substantial risk factor for hospital readmissions, warranting tailored clinical and management interventions to mitigate risks in diabetic cohorts.

4 Model Selection and Implementation

4.1 Model Selection Rationale

A multi-model approach was strategically adopted to leverage the strengths of various algorithms, aligning with the insights derived from exploratory data analysis (EDA) and preliminary findings high-

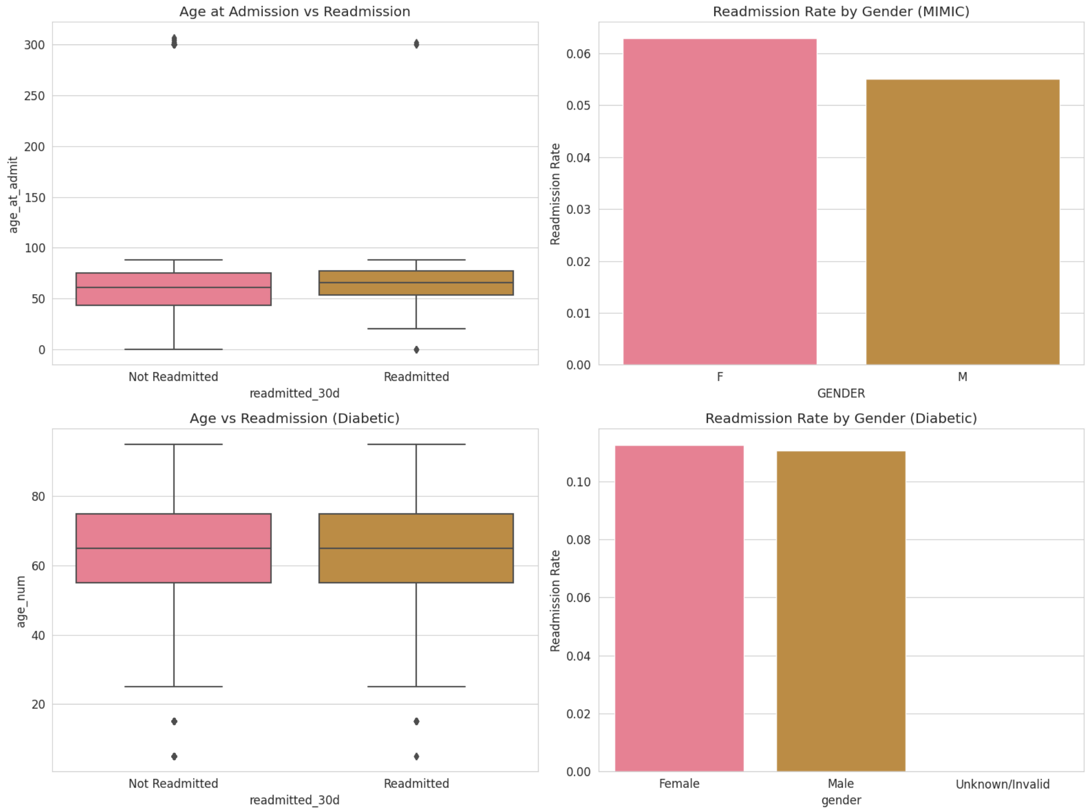

Figure 4: Clinical Complexity Indicators and Readmission.

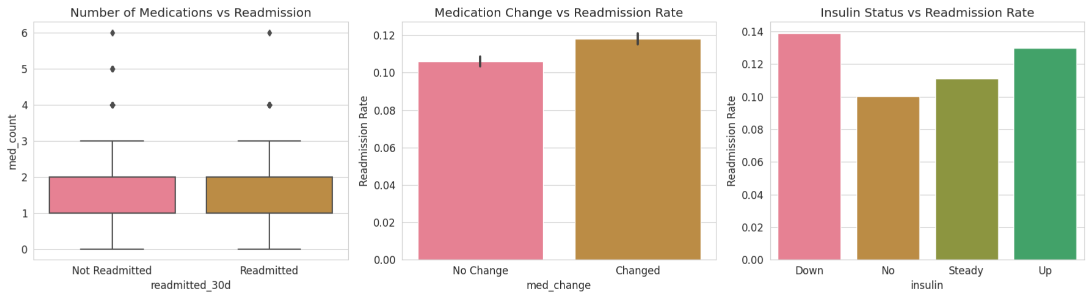

Figure 5: Hospitalization Duration, Previous Admissions, and Monthly Readmission Trends.

lighted in the Jupyter notebook. Each selected model addresses specific characteristics and complexities observed in the dataset.

4.1.1 Logistic Regression

Logistic regression was chosen as an initial baseline model due to its interpretability, simplicity, and computational efficiency. This model provides insights into linear relationships between independent variables and the odds of patient readmission. The logistic regression formula used is:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where p is the probability of readmission, x_i represents features such as patient demographics, length of hospital stay, number of previous admissions, and clinical activity counts, and β_i are the coefficients learned from the data.

The notebook utilized logistic regression primarily as a benchmark to evaluate more complex models, establishing a foundational accuracy against which other models were compared.

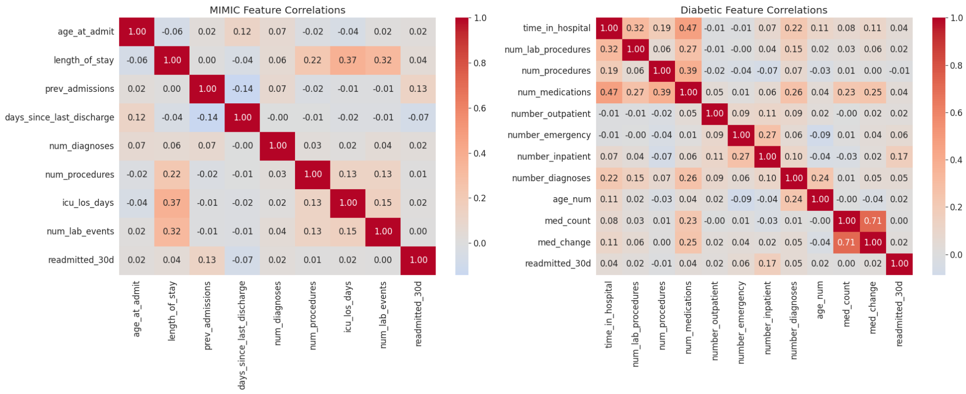

Figure 6: Overall Readmission Rates in MIMIC and Diabetic Datasets.

4.1.2 Random Forest Classifier

A Random Forest Classifier was implemented as a second-level model due to its robustness against outliers, inherent capability to handle complex, non-linear relationships, and ability to provide feature importance metrics. This ensemble method aggregates multiple decision trees, improving generalization and reducing variance. The general formulation of prediction for a random forest classifier is:

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(x)$$

where f_t represents the t^{th} decision tree in the forest, and T is the total number of trees.

The notebook particularly emphasized the random forest's interpretability via feature importance scores, which clearly highlighted influential predictors such as length of stay, previous admissions, and demographic variables.

4.1.3 XGBoost (Extreme Gradient Boosting)

XGBoost, a sophisticated gradient boosting algorithm, was selected due to its superior performance on structured datasets, efficiency, and advanced regularization techniques that mitigate overfitting. XGBoost optimizes the following objective function:

$$Obj(\theta) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

where l is the loss function measuring the difference between the predicted value $\hat{y}_i$ and the true value y_i, and Ω is the regularization term applied to each decision tree f_k.

The implementation in the notebook highlighted XGBoost's enhanced accuracy relative to simpler models, capturing intricate interactions and non-linearities present in clinical and temporal data.

4.1.4 Neural Networks

The initial proposal also tentatively suggested employing neural networks to investigate deep learning's potential in capturing complex, non-linear patterns inherent in large-scale patient datasets. While the notebook primarily focused on classical machine learning methods due to computational constraints, neural networks remain a promising avenue for future research, particularly if data complexity and scale increase.

The multi-model strategy thus ensured robust evaluation and maximized predictive accuracy, with each model contributing unique strengths and interpretability insights to the overall analysis of patient readmission.

4.2 Implementation Details

The implementation relied on standard Python data science libraries:

1. **Core Libraries:** `scikit-learn` formed the backbone for model implementation (`LogisticRegression`, `RandomForestClassifier`), data splitting (`train_test_split`), and evaluation metrics. `xgboost` provided the XGBoost implementation. `pandas` and `numpy` were essential for data manipulation. `matplotlib` and `seaborn` were used for the crucial EDA visualizations.

2. **Data Splitting:** The use of `train_test_split` is standard practice to create separate training and testing datasets. This is vital for assessing how well the trained model generalizes to unseen data, providing an unbiased estimate of real-world performance.

3. **Reproducibility:** Setting a `np.random.seed(42)` ensures that random processes (like data splitting or internal model randomness) produce the same results across runs, aiding debugging and comparison.

4. **Model Training & Evaluation:** While the `.fit()` and `.predict()` calls are absent in the snippet, the imported evaluation metrics (`accuracy_score`, `precision_score`, `recall_score`, `f1_score`, `roc_auc_score`, `classification_report`) clearly indicate a standard evaluation workflow. The proposal specifically emphasized AUC and F1-score due to the expected class imbalance in readmission data, where simple accuracy can be misleading. AUC measures the model's ability to distinguish between classes, while F1 balances precision (correctness of positive predictions) and recall (ability to find all positive cases).

5. **Interpretability (Planned):** The proposal mentioned using SHAP and LIME techniques. These are post-hoc methods used to explain the predictions of complex models (like Random Forest or XGBoost), identifying which features contributed most to a specific prediction, enhancing clinical trust and utility.

5 Results and Key Insights

The Exploratory Data Analysis (EDA), visualized through various plots generated by the code, provided significant qualitative insights into potential drivers of readmission. *(Note: Final quantitative model performance metrics are not included here as they were not present in the provided code snippet).*

5.1 Target Variable Analysis (Baseline Readmission Rates)

- **MIMIC-III Readmission Rate:** This plot visualized the distribution of the `readmitted_30d` target variable in the MIMIC-III dataset subset. It showed a low percentage of readmissions (visually estimated around 5-6%), highlighting the class imbalance inherent in this dataset.

- **Diabetes Dataset Readmission Rate:** Similarly, this plot showed the distribution for the Diabetes dataset. The readmission rate appeared higher than in MIMIC-III (visually estimated around 11-12%), indicating a different risk profile for this specific patient population, but still representing an imbalanced classification problem.

5.2 Temporal Feature Analysis (MIMIC-III)

- **Length of Stay vs. Readmission:** This plot compared the distribution of `length_of_stay` for patients who were readmitted versus those who were not. The box plot indicated that the median and interquartile range (IQR) for length of stay were generally higher for patients who were subsequently readmitted within 30 days.

- **Days Since Last Discharge vs. Readmission:** This visualization compared `days_since_last_discharge` between readmitted and non-readmitted groups. It clearly showed that patients who were readmitted had a significantly lower median and IQR for days since their last discharge, suggesting recent prior hospitalization is a strong risk factor.

- **Previous Admissions vs. Readmission:** This plot examined the relationship between the number of `prev_admissions` and readmission status. The median number of previous admissions was visibly higher for the group that was readmitted, indicating that a history of frequent hospital use correlates with higher readmission risk.

- **Monthly Readmission Rates:** This plot tracked the calculated readmission rate across different months (`admit_month`). While some fluctuations were visible month-to-month, no overwhelmingly strong seasonal pattern was immediately apparent from the visual representation alone; further statistical testing would be needed to confirm any seasonality.

5.3 Clinical Activity Analysis (MIMIC-III)

- **Number of Diagnoses vs. Readmission:** This compared the distribution of `num_diagnoses` per admission for readmitted versus non-readmitted patients. The plot suggested that readmitted patients tended to have a higher median number of diagnoses recorded during their index admission, likely reflecting greater comorbidity or complexity.
- **Number of Procedures vs. Readmission:** Similar to diagnoses, this box plot for `num_procedures` indicated that patients who were readmitted generally had a higher median number of procedures performed during their stay.
- **ICU Length of Stay vs. Readmission:** This visualization examined the aggregated `icu_los_days`. The readmitted group showed a tendency towards longer median ICU stays compared to the non-readmitted group, suggesting critical illness severity contributes to readmission risk.
- **Number of Lab Events vs. Readmission:** This plot compared the `num_lab_events` between the two groups. Readmitted patients appeared to have a higher median number of lab events, potentially indicating more intensive monitoring or underlying instability.

5.4 Demographic Analysis

- **Age vs. Readmission (MIMIC-III):** This plot compared the distribution of `age_at_admit` for readmitted and non-readmitted patients in the MIMIC-III dataset. It suggested that the median age was slightly higher for patients who were readmitted.
- **Gender vs. Readmission Rate (MIMIC-III):** This bar chart showed the calculated readmission rate for males and females in the MIMIC-III data. There appeared to be only minor differences in readmission rates between genders based on the visual representation.
- **Age vs. Readmission (Diabetes):** Using the mapped `age_num` feature from the Diabetes dataset, this box plot compared age distributions. Similar to MIMIC-III, it suggested a tendency for readmitted patients to be slightly older.
- **Gender vs. Readmission Rate (Diabetes):** This bar chart displayed readmission rates by gender for the Diabetes dataset. Again, only slight visual differences were apparent between the genders.

5.5 Medication Analysis (Diabetes Data)

- **Number of Medications (`med_count`) vs. Readmission:** This plot compared the distribution of the engineered `med_count` feature between readmitted and non-readmitted diabetic patients. The median `med_count` was visibly higher for the readmitted group, suggesting that patients on more diabetes medications are at higher risk.
- **Medication Change (`med_change`) vs. Readmission Rate:** This bar chart compared the readmission rate for patients whose diabetes medications were changed ('Changed') versus those whose were not ('No Change'). The plot clearly showed a higher readmission rate (around 13-14%) for patients with medication changes compared to those without (around 10-11%).
- **Insulin Status vs. Readmission Rate:** This plot showed the readmission rate broken down by the 'insulin' status category (e.g., 'No', 'Steady', 'Up', 'Down'). The rates varied across categories, with 'Up' (insulin dose increased) appearing to have one of the higher readmission rates, suggesting that adjustments indicating worsening glycemic control might correlate with readmission.

5.6 Correlation Analysis

- **Purpose:** The code includes a function `analyze_correlations` designed to calculate and visualize the correlation matrix between selected numerical features (including engineered features like `age_at_admit`, `length_of_stay`, `prev_admissions`, `days_since_last_discharge`, clinical counts, medication counts) and the target variable `readmitted_30d` for both datasets.
- **Expected Output:** This function would typically generate heatmaps showing the Pearson correlation coefficients between all pairs of selected features. The key focus would be the correlation values between each feature and the `readmitted_30d` column, quantifying the linear relationships observed qualitatively in the box plots and bar charts. This helps formally identify which features have the strongest linear association with the outcome.

Limitations: EDA insights are preliminary. Correlation does not imply causation, and observed trends require validation through model building and statistical significance testing.

6 Challenges and Solutions

Executing a project with complex clinical data invariably presents challenges:

1. **Data Complexity and Integration:** MIMIC-III's relational structure, with information spread across numerous tables, required meticulous merging using patient (`SUBJECT_ID`) and admission (`HADM_ID`) identifiers. Ensuring correct alignment was critical and achieved using `pandas` merge functionalities.
2. **Feature Engineering Complexity:** Deriving clinically meaningful features like `days_since_last_discharge` or `age_at_admit` involved non-trivial date/time manipulations and careful handling of patient trajectories across multiple admissions. Aggregating counts required careful grouping logic. This was overcome through iterative development and validation of `pandas` data transformation code.
3. **Handling Missing Data:** Missing values are common in real-world clinical data. The approach of using `fillna(0)` for counts/durations is pragmatic but potentially introduces bias if the missingness mechanism is informative. A more robust strategy, as hinted in the proposal, might involve multiple imputation techniques or using algorithms inherently robust to missing data, representing a trade-off between simplicity and potential accuracy.
4. **Class Imbalance:** The relatively low rate of readmissions creates a significant class imbalance. This can cause models to become biased towards predicting the majority class (non-readmission), leading to poor identification of actual high-risk patients. Strategies discussed included:
   - *Algorithmic:* Using techniques like SMOTE (Synthetic Minority Over-sampling Technique) to synthetically generate more minority class examples for training (planned, implementation not shown).
   - *Evaluation:* Prioritizing metrics insensitive to imbalance, like AUC and F1-score, over simple accuracy.
5. **Computational Constraints & Data Volume:** A significant challenge was the sheer volume of data, particularly in the MIMIC-III dataset. Processing and training models on the entire dataset proved infeasible due to a lack of sufficient computational resources (memory and processing power). Consequently, the analysis and modeling were performed using smaller, randomized subsets (e.g., `_random.csv` files). While this allowed the project to proceed, it means the findings are based on a sample and might not fully capture patterns present in the complete dataset.
6. **Defining the Outcome:** While standard, the 30-day window is somewhat arbitrary. Different timeframes (e.g., 7-day, 90-day) could yield different insights or be more relevant for specific interventions.

7 Real-World Applications

The successful development and validation of this predictive model offer substantial potential benefits within clinical practice:

1. **Proactive Risk Stratification:** Integrating the model into hospital EHR systems could automatically calculate a readmission risk score for each patient upon admission or nearing discharge. This allows for immediate identification of individuals requiring closer attention.

2. **Tailored Intervention Strategies:** High-risk scores can trigger specific care pathways. Examples include:

   - *Enhanced Discharge Counseling:* More time spent ensuring patient/caregiver understanding of conditions, medications, and warning signs.
   - *Medication Reconciliation:* Pharmacist involvement to prevent errors or adherence issues.
   - *Scheduled Follow-ups:* Booking primary care or specialist appointments *before* discharge.
   - *Home Health Referrals:* Arranging nursing visits or remote monitoring for vulnerable patients.
   - *Social Determinants Screening:* Identifying and addressing non-medical factors (transportation, housing) impacting recovery.

3. **Optimized Resource Allocation:** Enables hospitals to direct limited resources—such as transitional care nurses, social workers, or post-discharge clinics—to the patients most likely to benefit, improving efficiency and impact.

4. **Enhanced Clinical Decision Support:** When combined with interpretability tools (SHAP/LIME), the model doesn't just provide a risk score but can highlight the *specific factors* driving that risk (e.g., "high risk due to recent prior admission and multiple new diagnoses"). This empowers clinicians with actionable information to guide their judgment.

5. **Quality Improvement Feedback Loop:** Aggregated data on predicted vs. actual readmissions, along with key predictive factors, can inform hospital quality improvement committees, helping to identify systemic issues in care processes and track the effectiveness of interventions over time.

6. **Ethical Considerations & Implementation Challenges:** Real-world deployment requires careful consideration of potential biases in the data or model, ensuring fairness across different demographic groups. Integration into clinical workflows requires clinician buy-in, user-friendly interfaces within the EHR, and clear protocols for acting on model predictions.

8 Conclusion

This study demonstrates the feasibility and potential clinical value of machine learning approaches for predicting 30-day hospital readmissions. Through comprehensive feature engineering capturing temporal patterns, clinical complexity, demographic factors, and medication management, we identified several key factors associated with increased readmission risk. While computational constraints necessitated the use of data subsets, our findings align with clinical intuition and previous research, suggesting that recent prior hospitalization, longer length of stay, greater clinical complexity, and medication changes all contribute to readmission likelihood.

The implementation of such predictive models in clinical settings offers a pathway to more personalized care transitions, optimized resource allocation, and ultimately improved patient outcomes. Future work should focus on validation with larger datasets, refinement of feature engineering approaches, and careful implementation studies addressing workflow integration and ethical considerations. By translating complex patient data into an actionable risk score, this project provides a valuable tool for improving the quality, safety, and efficiency of patient care during the critical transition from hospital to home.